$\Delta(3n^2)$ and $\Delta(6n^2)$ in $SU(3)$

Coset formulas, coefficient proof, and matrix verification

Computational verification note

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Outcome. Both proposed formulas are correct. The standard 3-dimensional matrices were built for $n = 2, 3, 4, 5$ (orders 12–150). All 240 coefficients through total degree six agree with the exact Q_n/R_n formulas. Eight determinant-level comparisons have maximum error 8.89×10^{-16}.

1 Groups and master identity

Put $\eta = e^{2\pi i/n}$ and

$$N_n = \{D(a, b) = \operatorname{diag}(\eta^a, \eta^b, \eta^{-a-b}) : a, b \in \mathbb{Z}/n\}.$$

For any finite matrix group, the coefficient identity

$$[t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}] \frac{1}{|G|} \sum_g \prod_j \det(I - t_j g)^{-1} = \dim\Big(\bigotimes_s \operatorname{Sym}^{\tau_s} V\Big)^G \tag{1}$$

follows by expanding symmetric-power characters and taking the trace of the Reynolds projector.

Let P be the permutation matrix of a 3-cycle. Then $\Delta(3n^2) = N_n \rtimes \langle P\rangle$. For $\pi \in S_3$ use the determinant-one lift $\widetilde{P}_\pi = \operatorname{sgn}(\pi) P_\pi$; then $\Delta(6n^2) = N_n \rtimes \widetilde{S}_3$.

2 Coset proof

Define

$$\mathcal{A}_n = \frac{1}{n^2} \sum_{a,b} \prod_j \frac{1}{(1 - \eta^a t_j)(1 - \eta^b t_j)(1 - \eta^{-a-b} t_j)}. \tag{2}$$

For DP or DP^2, the three entries around the monomial 3-cycle multiply to $\det D = 1$, so $\det(I - tDP) = 1 - t^3$. Therefore

$$\boxed{\mathfrak{M}_{\Delta(3n^2)} = \tfrac{1}{3}\mathcal{A}_n + \tfrac{2}{3}\operatorname{Exp}_\sigma(p_3).} \tag{3}$$

For a transposition coset in the determinant-one lift, let α be the product around the 2-cycle and β the fixed-coordinate entry. Since the underlying permutation has determinant -1, $-\alpha\beta = 1$ and $\beta = -\alpha^{-1}$. Thus

$$\det(I - tg) = (1 - \alpha t^2)(1 + \alpha^{-1} t).$$

As $D(a, b)$ varies, α is uniform in μ_n. Put

$$\mathcal{B}_n = \frac{1}{n} \sum_{a \bmod n} \prod_j \frac{1}{(1 - \eta^a t_j^2)(1 + \eta^{-a} t_j)}. \tag{4}$$

The identity, 3-cycle, and transposition proportions in S_3 are 1/6, 1/3, and 1/2. Hence

$$\boxed{\mathfrak{M}_{\Delta(6n^2)} = \tfrac{1}{6}\mathcal{A}_n + \tfrac{1}{3}\operatorname{Exp}_\sigma(p_3) + \tfrac{1}{2}\mathcal{B}_n.} \tag{5}$$

3 Exact coefficient formulas

For $\tau = (\tau_1, \ldots, \tau_\ell)$, let $Q_n(\tau)$ count rows (x_s, y_s, z_s) with $x_s + y_s + z_s = \tau_s$ and $\sum_s x_s \equiv \sum_s y_s \equiv \sum_s z_s \pmod n$. Root-of-unity orthogonality gives $[m_\tau]\mathcal{A}_n = Q_n(\tau)$.

Let

$$R_n(\tau) = \sum_{\substack{x_s+2y_s=\tau_s \\ \sum y_s \equiv \sum x_s \ (n)}} (-1)^{\sum x_s}.$$

This follows by expanding $(1+\alpha^{-1}t)^{-1}$ and $(1-\alpha t^2)^{-1}$. Finally, $[m_\tau]\operatorname{Exp}_\sigma(p_3)$ is 1 exactly when every τ_s is divisible by 3. Therefore

$$[m_\tau]\mathfrak{M}_{\Delta(3n^2)} = \tfrac{1}{3}Q_n(\tau) + \tfrac{2}{3}\mathbf{1}_{3|\tau_s\ \forall s}, \tag{6}$$

$$[m_\tau]\mathfrak{M}_{\Delta(6n^2)} = \tfrac{1}{6}Q_n(\tau) + \tfrac{1}{3}\mathbf{1}_{3|\tau_s\ \forall s} + \tfrac{1}{2}R_n(\tau). \tag{7}$$

Their integrality is automatic from (1), even though (6)–(7) display rational weights.

4 Verification design and results

For each n, all matrices $D(a,b)P^k$ and $D(a,b)\widetilde{P}_\pi$ are constructed. The code checks group size and determinant one. It then compares: (i) eigenvalue-based symmetric-power character averages; (ii) exact enumeration of Q_n and R_n; and (iii) the direct determinant average at $(0.07, 0.11)$ against (3) or (5).

n	$\lvert\Delta(3n^2)\rvert$	$\lvert\Delta(6n^2)\rvert$	degree	checks/family	max. error	result
2	12	24	0–6	30	4.45×10^{-16}	**PASS**
3	27	54	0–6	30	6.67×10^{-16}	**PASS**
4	48	96	0–6	30	6.67×10^{-16}	**PASS**
5	75	150	0–6	30	8.89×10^{-16}	**PASS**

There are two families for each row, hence $2 \cdot 4 \cdot 30 = 240$ exact coefficient checks. As useful spot checks, for $n = 2$ the coefficients at $\tau = (3), (2,1), (1,1,1)$ are $(1,1,2)$ for $\Delta(12)$ and $(0,0,1)$ for $\Delta(24)$.

5 Novel diagnostic

The coset decomposition is also a debugging tool. A mismatch isolated to $\mathcal{B}_n$ almost always signals the missed minus sign in the determinant-one lift of a transposition. A mismatch isolated to $\operatorname{Exp}_\sigma(p_3)$ signals an incorrect monomial-cycle product. This three-channel comparison is stronger than a single ordinary Molien specialization.

Reproduction

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.venv/bin/python -m \
  for_this_guy.generalized_molien_extensions.delta_su3.verification
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